CHAPTER 8

Radiation Effects on Matter

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Soon after the discoveries of X-rays and radioactivity it was learned that radiation could cause changes in matter. In 1901 P. Curie found that when a radium source was placed on his skin, wounds were produced that were difficult to heal. In 1902 skin cancer was shown to be caused by the radioactivity from radium but 5 years later it was learnt that radium therapy could be used to heal the disease. Large radiation doses were found to kill fungi and microorganisms and produce mutations in plants.

Glass ampoules containing milligrams of radium darkened within a few months and became severely cracked, allowing the leakage of radon gas. In the early years of the investigation of radioactivity, emphasis was on radium and its decay products. Among the radiation effects observed were the fluorescence induced in different salts and the changes in their crystallographic form resulting in colour changes from delocalised charges in the crystals. Metals were found to lose their elasticity and become brittle due to delocalised atoms breaking the inherent structure. Radiation was also found to have a profound effect on the chemical composition of solutions and gases. Water, ammonia and simple organic substances decomposed into more elementary constituents and also combined

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into more complex polymeric products. Radiation decomposition (*radiolysis*) of water caused evolution of hydrogen and oxygen gas and formation of hydrogen peroxide. Conversely, it was shown that water could be synthesized through irradiation of a mixture of H_2 and O_2. In 1911 S. Lind found that 1 g of radium exposed to air resulted in the production of 0.7 g of ozone per hour. By relating this *radiation yield* to the number of ions produced by the amount of radiation, Lind initiated the quantitative treatment of radiation-induced changes.

8.1. ENERGY TRANSFER

The chemical effects of radiation depend on the composition of matter and the amount of energy deposited by the radiation. In this section we consider only the energy transfer. For this purpose it is practical to divide high energy radiation into (1) charged particles (e^-, e^+, α, etc) and (2) uncharged particles (n) and electromagnetic radiation (γ). The latter produce recoil atomic ions, products of nuclear reactions and electrons as charged secondary ionizing particles. The terms direct and indirect ionizing radiation are often used for (1) and (2) respectively.

The amount of energy imparted to matter in a given volume is

$$E_{\text{imp}} = E_{\text{in}} + \Sigma Q - E_{\text{out}} \tag{8.1}$$

where E_{in} is the energy (excluding mass energy) of the radiation entering the volume, E_{out} is the energy of the radiation leaving the volume, ΣQ is the sum of all Q-values for nuclear transformation that have occurred in the volume. For a beam of charged particles $E_{\text{in}} = E_{\text{kin}}$; for γ-rays it is E_γ. If no nuclear transformations occur, $\Sigma Q = 0$. For neutrons which are captured and for radionuclides which decay in the absorber, $\Sigma Q > 0$; in the case of radionuclides already present in the absorber, $E_{\text{in}} = 0$.

8.1.1. Charged particles

We learned in the previous chapter that the energy of charged particles is absorbed mainly through ionization and atomic excitation. For positrons the annihilation process (at $E_{\text{kin}} \approx 0$) must be considered. For electrons of high kinetic energy bremsstrahlung must be taken into account. However, in the following text we simplify by neglecting annihilation and bremsstrahlung processes. The bremsstrahlung correction can be made with the aid of Figure 7.9 which gives the *average specific energy loss* of electrons through ionization and bremsstrahlung.

It has been found that the average energy, w, for the formation of an ion pair in gaseous material by charged particles is between 25 and 40 eV. For the same absorbing material it is fairly independent of the type of radiation and the energy. Table 8.1 lists values of w in some gases. The ionization potentials j of the gases are lower than the